

TECHNICAL DEEP-DIVE

Inaya Ecosystem — Developer Reference

Contract functions, API routes, data models, and SDK signatures — a lookup reference grounded directly in the real code.

Document INAYA-DEV-2026-V1 · Classification Internal · September 2026

SECTION 01

Repo & Entry-Point Map

PATH	ENTRY POINT
contracts/ (repo root)	*.sol — Hardhat project, hardhat.config.js at repo root
scripts/ (repo root)	deploy*.cjs / deploy*.js, verify-*.cjs, simulate-security-nodes.js
inaya-network-dapp/	Next.js 14 App Router — src/app/, src/lib/, src/app/api/
inaya-network-dapp/custody-sdk/	src/index.js (InayaKernel), src/crypto.js, src/contracts.js
inaya-network-dapp/custody-sdk/packages/node-daemon/	bin/inaya-node-daemon.js (commander CLI)
src/lib/chain-adapters/ (bridge + interop)	registry.js (capability model), interop/ (Wormhole)
src/lib/ai-action-requests.js, auditChain.js	AI guarded-execution state machine + cryptographic audit trail
inaya-mobile/	Expo — App.js, src/screens/, src/utlis/, src/providers/
inaya-desktop/	src-tauri/src/lib.rs, tauri.conf.json
inaya-dapp-desktop/	src-tauri/src/lib.rs, tauri.conf.json

SECTION 02

Contract Function Reference

InayaStaking.sol

Constructor(stakingToken, rewardToken) — both = \$INAYA address.

- stake(uint256 amount, uint256 lockPeriodDays) — lockPeriodDays ∈ {0,30,90}, multiplier 1.00x/1.25x/1.50x, locked in on first stake
- withdraw(uint256 amount) — reverts if block.timestamp < lockExpiry
- claimReward() / exit() (withdraw all + claim)
- earned(address) / rewardPerToken() / getUserTier() → "None"/"Standard"/"Enterprise Priority"
- owner-only: setRewardRate, fundRewardPool, setEnterpriseTierThreshold, recoverForeignToken (blocked on staking/reward token)
- events: Staked, Withdrawn, RewardPaid, RewardRateUpdated, RewardPoolFunded

InayaEgressTimelockVault.sol

Constructor(inayaToken, usdtToken, stakingContract, operationalTreasury).

- executeSemiAnnualHarvest() — public, callable by anyone once currentEpochStart + 180 days has passed; sweeps full INAYA balance to stakingContract via raw transfer (not fundRewardPool)
- forwardUSDT() — public, sweeps full USDT balance to operationalTreasury
- getNextHarvestCountdown() view
- owner-only: setStakingContract, setOperationalTreasury

InayaCorporateEscrow.sol

Constructor(usdtToken) only.

- createEscrow(address corporate, address node, uint256 totalAmount) — safeTransferFrom(caller) into escrow, MONTHS=12 fixed schedule
- releaseMonthlyPayout(uint256 scheduleId) — public, callable by anyone once due
- owner-only: setMonthSeconds (testnet-only timing override), reassignNode
- events: EscrowCreated, MonthlyReleased, EscrowClosed

InayaNodeRegistry.sol

Constructor(usdtToken, verifierWallet). SCOPE NOTE in header: metrics are coordinator-verified, NOT cryptographic proof-of-storage.

- registerNode(uint256 capacityGB) — self-serve
- updateNodeMetrics(...) — verifier-only
- queueSettlement / queueSettlementsBatch — verifier-only, computes commission (30/40/50% by Entry/Mid/Enterprise tier), does not move funds
- releaseSettlement / releaseSettlementsBatch — public, callable by anyone, only after SETTLEMENT_DELAY = 36 hours
- owner-only: setVerifierWallet, setTierThresholds, withdrawExcessReserve

InayaProofRegistry.sol

Constructor(custodyAddress) — immutable IInayaCustody reference.

- registerMerkleRoot(bytes32 fileHash, bytes32 merkleRoot, uint256 chunkCount, address node) — requires msg.sender == custody.assets(fileHash).owner; node must be pre-approved via setNodeRegistered
- verifyChunkProof(...) — onlyOwner today (backend-checked); header comment documents a planned permissionless + slashing path
- events: MerkleRootRegistered, ProofVerified, NodeRegistrationChanged

InayaCustody (interface only — IInayaCustody)

Deployed, source not tracked in this repo. Interface as referenced from InayaProofRegistry:

- assets(bytes32 fileHash) view → {owner, cidAlpha, cidBeta, size, timestamp}
- batchRegisterAssets(bytes32[] fileHashes, uint256[] fileSizes, string[] shardACIDs, string[] shardBCIDs)
- event AssetRegistered

RevenueRouter (interface only)

Deployed, source not tracked in this repo. Called via inline ABI from page.js and the Stripe webhook route:

- processCorporateInvoice(uint256 usdtAmount) external

Security Layer contracts

InayaThreatRegistry (status ledger, updated only by Reporter), InayaThreatReporter (confirmThreat(bytes32 threatId, uint8 category, uint16 confidenceBps, bytes32 contributingNodesHash) — relayer-only), InayaNodeReputation (checkpointReputation(...) — relayer-only, periodic), InayaSecurityPolicy (publishPolicy(uint256 version, bytes32 policyHash, string policyURI) — owner/relayer-only).

Oracle & Automation Layer contracts (deployed to BSC Testnet)

InayaOracleRegistry (registerSource(bytes32 id, string dataType, address submitter, uint256 updateFrequency) — owner-only; isAuthorizedSubmitter(id, address) view), InayaOracleAdapter (submitData(bytes32 id, uint256 value, uint256 reportedTimestamp) — reverts on future/stale timestamp, sub-minimum-interval, or excess deviation; getLatestData(id) / isStale(id) views), InayaAutomationRegistry (registerTask(bytes32 id, address target, bytes4 selector, string condition) — owner-only; recordExecution(id, bool success, uint256 nextEligible, bytes32 txHash) — worker/owner-only, pure audit-trail write, never forwards a call to target).

SECTION 03

custody-sdk (InayaKernel) API Reference

FUNCTION	PURPOSE
<code>generateSecureSalt()</code>	16-byte random salt via <code>crypto.getRandomValues</code>
<code>deriveVaultKey({passkey, salt, iterations=100000, algo})</code>	PBKDF2 → 32-byte AES key
<code>disperseAndSlice({file, encryptionKey})</code>	Encrypt (AES-GCM-256) + base64-midpoint-split into <code>shardAlpha/shardBeta</code>
<code>anchorToLedger(...)</code>	Calls <code>InayaCustody.batchRegisterAssets()</code> with CIDs (not raw shard bytes)
<code>retrieveAndReconstruct({connection, fileHash, passkey, fetchShard})</code>	Reads <code>assets(fileHash)</code> , fetches both shards, reassembles
<code>reconstructAndDecrypt({shardAlpha, shardBeta, passkey})</code>	Concat → split salt/IV/ciphertext → re-derive key → AES-GCM decrypt
<code>encryptForPublicKey / decryptWithSecretKey / deriveEncryptionKeypairFromSignature</code>	X25519 + HKDF-SHA256 + XChaCha20-Poly1305 sealed-box — re-wraps the passkey for sharing, not the file
<code>connectWallet, approveFeeTokens</code>	ethers v6 wallet/allowance helpers
<code>Staking.{stake,unstake,claimReward,calculateReward,getStakedBalance}</code>	Thin wrapper over <code>InayaStaking</code>
Payments, Metadata, Analytics, events (<code>EventEmitter</code>)	Supporting modules
<code>errors.</code> <code>{InayaError,InayaValidationError,InayaWalletError,InayaContractError,InayaNetworkError}</code>	Typed error classes

Crypto primitives are `@noble/hashes`, `@noble/ciphers`, `@noble/curves` (pure JS) rather than `crypto.subtle` — chosen specifically for React Native compatibility, per an inline comment in `crypto.js`.

Release verification (v1.0.10-beta+)

Every git tag matching `v*` on `custody-sdk` (`.github/workflows/release.yml`) runs `npm test` → computes checksums → pins to IPFS (Pinata, non-blocking) → `npm publish --provenance` → commits `CHECKSUMS.md` → attaches the tarball to a GitHub Release. Reproduce independently per `docs/VERIFYING_RELEASES.md`:

- `git rev-parse <tag> ^{tree}` — compare to `CHECKSUMS.md`'s `git-tree-hash`. NOT `git archive | sha256sum` (tried first, found non-reproducible across git versions — see Section 05 of `ecosystem-architecture.js`).
- Download the `.tgz` attached to the GitHub Release, `sha256sum` it, compare to `CHECKSUMS.md`'s `npm-tarball-sha256` — always matches, since you're hashing the literal file CI attached rather than re-deriving it (npm pack tarball bytes aren't guaranteed identical across npm versions, even when file contents are).
- `npm view @inaya-network/custody-sdk@<version> dist.integrity / dist.shasum` — the registry's own recorded hash for the published tarball.
- `test/webCryptoCompat.test.mjs` — the committed proof that this package's crypto and the dApp's former inline `crypto.subtle` implementation are byte-identical and cross-decryptable.

SECTION 04

node-daemon CLI Reference

COMMAND	EFFECT
login	Reads INAYA_PRIVATE_KEY (env or prompt), encrypts at rest (PBKDF2-SHA256 200k iter + AES-GCM-256), writes ~/.inaya/node-daemon/config.json
register <capacityGB>	InayaNodeRegistry.registerNode(capacityGB) on-chain + POST /api/nodes/register off-chain
start	Foreground 5-min heartbeat loop → POST /api/nodes/heartbeat with {nodeId, operatorWallet, totalCapacityGB, usedCapacityGB:0, shardsStored:0} (last two hardcoded 0 — no shard-storage capability exists)
report <indicator> - -category <cat>	Security Layer signed observation → POST /api/security/report
service install / uninstall	Windows background service via node-windows

package.json description states it plainly: registers your node on-chain and reports heartbeat/telemetry. Does not store or serve shards. NODE_REGISTRY_ABI in this package has exactly two functions (registerNode, nodes) — no stake/settlement/withdraw calls exist in the daemon's on-chain surface.

SECTION 05

Backend lib/ Reference

Every file below follows the same shape: getXCollections(), ensureXIndexes() (module-level indexesEnsured guard), validateXInput() (throws, fail-closed).

FILE	DOMAIN
security.js	Threat reports/threats/policy/events/reputation-cache, computeThreatConfidence, getPublicSecurityStats
ai-security-tools.js	Security Assistant tool declarations + dispatcher
learn.js / learnConfig.js / youtube.js	Learn saved/progress/search-cache/video-cache, category+collection config, cached YouTube Data API client
ai-learn-tools.js	Learn Tutor tool declarations + dispatcher
orgs.js / orgPlans.js	Business Workspace auth (sessions, magic-links), org/plan definitions; canManageOrg/canAccessDepartment plus the additive canManageFinance/canAccessFinance/canManageHR/canAccessHR/isDepartmentManager/isSelfEmployeeRecord gates
document-workflow.js / document-permissions.js	Document state machine + activity log; access resolution + share tokens + getAccessibleScope() (the one org-wide 'what can this member see' resolver, extended per module as each shipped)
ai-business-tools.js	Business Assistant tool declarations + dispatcher (permission-aware) — list_documents/tasks/contacts/deals/suppliers/purchase_orders/products/invoices/expenses/employees, find_employee_document
task-workflow.js / deal-workflow.js / purchase-request-workflow.js / purchase-order-workflow.js	Business Operations transition-table workflows — {from,to,requiresManage,activityAction} pattern, one findOneAndUpdate per transition
inventory.js	recordStockMovement(), getStockLevel(), totalStockForProduct(), isLowStock() — stock_levels is \$inc-only, stock_movements is the append-only ledger
invoice-workflow.js / expense-workflow.js	transitionInvoice() (send/markPaid/cancel + cron-only markOverdueInvoices()), transitionExpense() (submit/approve/reject/cancel)
employee-workflow.js / leave-workflow.js	transitionEmployee() (onboarding→active→on_leave→terminated), transitionLeaveRequest(), getLeaveBalance() (computed fresh, never a stored counter)
attachments.js	createAttachment()/listAttachmentsForRecord()/serializeAttachment() — shared metadata layer for Finance receipts (EXPENSE) and HR documents (EMPLOYEE), same encrypt/shard/pin client pipeline as org_documents
org-activity-log.js	logOrgActivity()/listOrgActivityForRecord() — one domain-generic audit log every Business Operations/Finance/HR workflow writes into
rag/collections.js, chunking.js, embeddings.js, sanitize.js, vectorStore.js, retrieve.js, ingest.js, youtubeTranscript.js, queryCache.js, metrics.js	RAG pipeline — see Section 11 for function-level detail
rag/sources/{docsSources,securitySources,learnSources}.js	Per-domain source adapters feeding ingest.js — DOCS_SOURCES, SECURITY_SOURCES, LEARN_STATIC_SOURCES + ensureVideoTranscriptIngested()
dataroom.js	Investor Data Room visitors/sessions/documents/views
activity.js	DAU/WAU ping recording + stats
watcherPioneer.js	Watcher Pioneer enrollment/qualification
feedback.js	Bug reports / idea submissions
metadata-auth.js	Shared signed-message verification helper (verifyMetadataAuth)
email.js	Resend-backed transactional email (magic links, Data Room verify)

SECTION 06

API Route Reference (by domain)

DOMAIN	ROUTES
Security (public)	GET /api/security/{threat,feed,stats,policy}, POST /api/security/report, GET /api/security/reputation/[address], POST /api/security/events, GET /api/security/events
Security (admin)	GET /api/admin/security/{threats,nodes}, POST /api/admin/security/threats/[id]/override, GET POST /api/admin/security/policy
Security (AI)	POST /api/ai/security-chat
Security (cron)	GET /api/security/cron/checkpoint-reputation (CRON_SECRET-gated)
Learn	GET /api/learn/config, GET /api/learn/search, GET /api/learn/video/[videoid], GET POST /api/learn/saved, DELETE /api/learn/saved/[videoid], GET POST /api/learn/progress, POST /api/learn/report, POST /api/learn/analytics
Learn (AI)	POST /api/ai/learn-chat
Business Workspace (auth)	POST /api/orgs/create, /api/orgs/login/google, magic-link login/verify
Business Workspace (org)	department/project CRUD, document upload/list/retrieve/transition, permissions grant/update/revoke, share create/list/revoke, GET /api/orgs/share/[token] (public resolve)
Business Workspace (billing)	GET /api/orgs/billing, POST /api/orgs/billing/checkout, POST /api/orgs/billing/portal, GET /api/orgs/billing/plans
Business Workspace (AI)	POST /api/ai/business-chat
Business Workspace (tasks/CRM/procurement/inventory)	/api/orgs/tasks/**, /api/orgs/crm/{contacts,deals}/**, /api/orgs/procurement/{suppliers,requests,orders}/**, /api/orgs/inventory/{products,warehouses,movements}/**, GET /api/orgs/dashboard (aggregate summaries per module)
Business Workspace (finance)	/api/orgs/finance/invoices/** (incl. [id]/transition, [id]/activity), /api/orgs/finance/expenses/** (incl. [id]/attachments, [id]/transition), /api/orgs/finance/payments/** (incl. [id]/approve), GET /api/orgs/finance/reports (?type=revenue expenses outstanding paid-unpaid&format=json csv)
Business Workspace (HR)	/api/orgs/hr/employees/** (incl. [id]/attachments, [id]/leave-balance, [id]/transition, [id]/activity), /api/orgs/hr/leave-requests/** (incl. [id]/transition), POST DELETE /api/orgs/hr/departments/[id]/manager
Business Workspace (finance/HR cron)	GET /api/cron/invoices-mark-overdue (CRON_SECRET-gated, nightly SENT→OVERDUE flip)
RAG (public, via existing assistants)	Consumed inside POST /api/ai/chat (Docs), /api/ai/security-chat, /api/ai/learn-chat — no separate public RAG endpoint
RAG (admin)	GET /api/admin/rag/stats, POST /api/admin/rag/reingest
RAG (cron)	GET /api/cron/rag-reingest (CRON_SECRET-gated)
Data Room (public)	POST /api/dataroom/request-access, GET /api/dataroom/verify, GET /api/dataroom/session, POST /api/dataroom/accept-nda, GET /api/dataroom/documents, GET /api/dataroom/documents/[id]/stream, POST /api/dataroom/documents/[id]/view-event
Data Room (admin)	GET POST /api/admin/dataroom/documents, DELETE /api/admin/dataroom/documents/[id], GET /api/admin/dataroom/visitors, POST /api/admin/dataroom/visitors/[id]/revoke

DOMAIN	ROUTES
Nodes	POST /api/nodes/register, POST /api/nodes/heartbeat, GET /api/nodes/settlements/release (cron)
Referrals	POST /api/referrals/{activate,initiate,webhook}, GET /api/referrals/{status,history,leaderboard}
Watcher Pioneer	POST /api/watcher/{enroll,qualify}, GET /api/watcher/status
Activity	POST /api/activity/ping, GET /api/admin/activity
Other	POST /api/faucet, POST /api/stripe-webhook, POST /api/create-payg-checkout-session, POST /api/feedback/submit, POST /api/feedback/upload, GET /api/admin/feedback

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SECTION 07

MongoDB Collections Reference

COLLECTION GROUP	KEY COLLECTIONS
Security	security_reports, security_threats, security_policy, security_events, security_reputation_cache
Learn	learn_saved, learn_progress, learn_search_cache (TTL 24h), learn_video_cache (TTL 24h), learn_reports, learn_analytics_events
Business Workspace (core)	orgs, org_members, sessions, magic_links, departments, projects, documents, document_permissions, document_shares, document_activity, project_members
Business Operations	tasks, crm_contacts, crm_deals, suppliers, purchase_requests, purchase_orders, warehouses, products, stock_levels, stock_movements, org_activity (one shared domain-generic audit log for every module below, not just Business Operations)
Finance & HR	invoices, expenses, payments, employees, leave_requests, attachments (shared metadata store — EXPENSE and EMPLOYEE relatedRecordType)
RAG	rag_chunks (with Atlas Vector Search + Atlas Search indexes), rag_sources, rag_ingestion_runs, rag_query_log, rag_embedding_cache
Data Room	dataroom_documents, dataroom_visitors, dataroom_magic_links, dataroom_sessions, dataroom_views
Activity	activity_pings — one doc per {surface, identityId, date}, unique compound index
Referrals / Watcher Pioneer / Feedback	referrals, watcher_enrollments, feedback_submissions (each with their own lib.js as listed in Section 05)

SECTION 08

AI Assistant Tool-Calling Pattern

Identical shape across all four assistants (src/app/api/ai/{chat,business-chat,security-chat,learn-chat}/route.js)
— chat/route.js is the Docs Assistant, the other three are named for their domain:

- Model: gemini-flash-latest, thinkingConfig: { thinkingLevel: "low" }, maxOutputTokens: 800
- MAX_TOOL_ROUNDS = 5 — a bounded loop of generateContent → check functionCalls → run tools → feed functionResponse back in
- generateContentWithRetry() — retries on HTTP 429/503 with a [700ms, 1800ms] backoff schedule
- Route-level: maxDuration = 90, not streaming (deliberate — unpredictable number of tool round-trips before a final answer)
- Per-assistant: buildXContext() computed once per request, X_TOOL_DECLARATIONS (Gemini Type.OBJECT schemas), runXTool(name, args, ctx) dispatcher, xSystemInstruction() builder

SECTION 09

Auth Patterns — Technical Detail

PATTERN	IMPLEMENTATION DETAIL
Wallet signed-message	ethers.verifyMessage(reconstructedMessage, signature) === expectedAddress, 5-minute timestamp freshness window, reimplemented per-feature (metadata-auth.js has the canonical version, security.js/watcherPioneer.js reimplement locally by convention)
Magic-link	generateToken() (random), hashToken() (stored hashed, never plaintext), TTL-indexed Mongo collection, consumeLoginToken() atomic findOneAndUpdate to prevent replay
Session (web)	HttpOnly cookie, createSession(email) helper in orgs.js
Session (mobile)	Bearer token, same session concept, stored via expo-secure-store
Share tokens	generateShareToken()/hashShareToken(), atomic findOneAndUpdate-based single-consume pattern (document-permissions.js), reused as a template for Data Room's magic-link tokens

SECTION 10

Testing & Verification Conventions

- Hardhat tests: `test/security-layer.test.js` (17 tests — access control, confirm/re-confirm, false-positive override, 4-signer simulation)
- Node backend tests: `test/*.test.mjs` run via `node --test` against a real MongoDB instance (not mocked) — `security.test.mjs` (11), `learn.test.mjs`, `dataroom.test.mjs`, `activity.test.mjs`, `feedback.test.mjs`, `watcher-pioneer.test.mjs` — 140/140 passing as of the Security Layer build, no regressions
- Business Operations: `business-operations-tasks.test.mjs`, `crm-workflow.test.mjs`, `procurement-workflow.test.mjs` (incl. real inventory-ledger integration — a receipt genuinely moves stock), `document-permissions.test.mjs`
- Finance & HR: `finance-workflow.test.mjs` (8 tests — invoice lifecycle, cron overdue-flip + idempotency, org isolation, expense approval gate, payment record→approve replay-safety), `hr-workflow.test.mjs` (9 tests — employee lifecycle, leave approval + computed-balance correctness, self-access vs. HR-role vs. Department Manager scope) — both caught real bugs before shipping: a race condition in PO receiving (stock movements were applying before the PO's own optimistic-concurrency-guarded update), and a Department Manager visibility bug in `getAccessibleScope()` (Section 12)
- RAG: `rag-ingestion.test.mjs`, `rag-security.test.mjs`, `rag-attribution.test.mjs`, `rag-learn-transcript.test.mjs`, `rag-retrieval.test.mjs` (28 tests total) — run against the real Atlas cluster and real Gemini embeddings, not mocked; the relevance threshold (Section 11) was empirically calibrated from live measurements taken during this test work, not guessed
- On-chain simulation: `scripts/simulate-security-nodes.js` — 4 throwaway signers report a clearly-fake test indicator end-to-end through to a real on-chain `confirmThreat()` transaction on BSC Testnet
- Manual browser verification is the norm for UI work in this codebase — `curl/fetch` checks for API correctness, then a real browser pass (or explicit statement that one hasn't happened yet) before anything is called "verified"

SECTION 11

RAG Pipeline — Function Reference

FUNCTION	FILE	PURPOSE
chunkMarkdownByHeading() / chunkParagraphs() / chunkStructuredSections() / chunkQaPairs()	rag/chunking.js	Source-shape-specific chunkers — structured-sections for fundraising-docs' {sections: [[blocks]]} shape, qa-pairs for FAQ arrays, heading-split for markdown, paragraph-split as the fallback
embedText() / embedChunkText() / embedQueryText()	rag/embeddings.js	Wraps Gemini's models.embedContent (gemini-embedding-001, 768 dims) — RETRIEVAL_DOCUMENT taskType for ingestion, RETRIEVAL_QUERY for search; content-hash-cached in rag_embedding_cache
sanitizeChunkText() / wrapContextBlock()	rag/sanitize.js	Injection-pattern neutralization before a chunk is ever fed into a model prompt, and delimited wrapping so retrieved context can't be confused with the system prompt
ensureIndexes() / upsertChunks() / deleteBySource() / deleteObsoleteChunks() / hybridSearch()	rag/vectorStore.js	Idempotent index creation (\$vectorSearch + \$search via collection.createSearchIndex()); hybridSearch() runs both queries and merges via Reciprocal Rank Fusion in application code
retrieveContext() / formatAttribution()	rag/retrieve.js	The one function every assistant calls. DEFAULT_MIN_RELEVANCE = 0.8 — hasResults gates purely on top vector score, no keyword-match override (removed after it let one irrelevant query pass)
ingestSource() / ingestAllStaticSources() / deleteSource()	rag/ingest.js	Content-hash diffing per source: insert new chunks, update changed ones, skip unchanged ones, remove chunks whose source content no longer produces them
fetchYouTubeTranscript(videoid)	rag/youtubeTranscript.js	Scrapes the public timedtext caption-track URL out of a video's watch-page HTML — unofficial (the official Data API's captions.download needs OAuth as the channel owner), always fails gracefully to null, never throws
recordRetrieval() / recordIngestionRun()	rag/metrics.js	Writes to rag_query_log / rag_ingestion_runs — fail-open, a metrics write never blocks the actual retrieval/ingestion it's recording

DOCS_SOURCES (rag/sources/docsSources.js) covers inaya-knowledge.js, business-workspace-guide.md, the FAQ page's faqs export, and all 15 scripts/fundraising-docs/content/*.js files — explicitly excluding custody-sdk's own docs (a nested, separately-git-excluded repo — see Section 20's build-breakage history for why touching it at all is risky) and the legacy KNOWLEDGE_ARTICLES collection.

SECTION 12

Business Operations & Finance/HR — Workflow Reference

Every transition below follows the same shape: one `findOneAndUpdate({..., status: fromState}, {$set: {status: toState}})` — a mismatched current status is a clean 409, no extra locking needed.

MODULE	TRANSITIONS
task-workflow.js	submit: DRAFT→IN_PROGRESS, complete: IN_PROGRESS→DONE, reopen: DONE→IN_PROGRESS, cancel: *→CANCELLED
deal-workflow.js	stage transitions across the pipeline, terminal: →WON / →LOST
purchase-request-workflow.js	submit: DRAFT→PENDING_APPROVAL, approve/reject: PENDING_APPROVAL→APPROVED REJECTED (requiresManage), cancel: DRAFT PENDING_APPROVAL→CANCELLED
purchase-order-workflow.js	submit→approve→order→receive (partial and full) — <code>receivePurchaseOrder()</code> applies inventory stock movements only AFTER the PO's own optimistic-concurrency-guarded status update succeeds (reordered after a real race condition was caught — see Section 10)
invoice-workflow.js	send: DRAFT→SENT, markPaid: SENT OVERDUE→PAID, cancel: DRAFT SENT OVERDUE→CANCELLED, markOverdueInvoices() (cron-only): SENT→OVERDUE
expense-workflow.js	submit: DRAFT→PENDING_APPROVAL, approve/reject: PENDING_APPROVAL→APPROVED REJECTED (requiresManage: <code>canManageFinance</code>), cancel: DRAFT PENDING_APPROVAL→CANCELLED
employee-workflow.js	activate: ONBOARDING→ACTIVE, placeOnLeave: ACTIVE→ON_LEAVE, returnFromLeave: ON_LEAVE→ACTIVE, terminate: ACTIVE ON_LEAVE→TERMINATED (requiresManage: <code>canManageHR</code>)
leave-workflow.js	approve/reject: PENDING→APPROVED REJECTED (requiresManage: <code>canManageHR</code>), cancel: PENDING→CANCELLED (requester or <code>canManageHR</code>); <code>getLeaveBalance()</code> = <code>annualLeaveAllocationDays</code> – <code>sum(this year's APPROVED day-spans)</code> , computed on every call

getAccessibleScope() extension pattern. Each module adds its own `visibleX` array (`visibleTasks`, `visibleContacts`, `visibleInvoices`, `visibleEmployees`, ...) to the same resolver rather than each having a separate scope function. Finance's arrays are gated by `canAccessFinance` on top of department scope; HR's `visibleEmployees` unions HR-scoped employees, Department-Manager-scoped employees (`managedDepartmentIds` — NOT filtered against the caller's own already-visible departments, a real bug fixed during this layer's testing, see the founder architecture doc's Section 24), and the caller's own self-record, deduplicated by `_id`.

SECTION 13

Multi-Chain Bridge & Interop Reference (September 2026)

FUNCTION / CONTRACT	PURPOSE
InayaTokenBridgeHome.bridgeOut(...)	BSC-side lock/burn entry point, source of every cross-chain transfer
InayaTokenBridgeSpoke.executeMessage(...) / receive_message(...)	Remote-chain entry point (EVM name / Solana-Move name) — validates threshold validator signatures, mints/releases on the destination
InayaChainRegistry.setSpokeBridgeAddress() / registerRemoteChain() / setTrustedRemoteContract()	Owner-only wiring — a chain has no real route until all three are called for it, not just deployed
listChainCapabilities() / getChainCapability(chainId)	src/lib/chain-adapters/registry.js — the single source a UI reads before ever describing a chain as 'supported'; SUPPORT_LEVELS run DISCOVERED(0) → READ_ONLY → WALLET → MESSAGE → TOKEN_TRANSFER → STAKING → FULL_ECOSYSTEM(6), each backfilled only from what's actually been verified
WormholeProvider.js	Wraps Wormhole's WTT/lock-and-mint flow; chain-id mapping is the thing that broke first (Sepolia/testnet IDs differ from their mainnet counterparts unlike BSC/Avalanche)
GET/POST /api/interop/wtt/{initiate,relay,status}	Relay cron (* / 5 min) fetches the Guardian-signed VAA via a plain REST call to Wormholescan's public API (not the @wormhole-foundation/sdk package — that broke the Next.js build, see Section 26 of the architecture doc) and completes the transfer via Inaya's relayer wallet

Per-chain proof records (real tx hashes both sides):
deployments/bridge/{avalancheFuji,solanaDevnet,hederaTestnet,aptosTestnet,suiTestnet}.json;
deployments/interop/wormhole-wtt/bscTestnet-attestation.json.

SECTION 14

Storage Backup & Redundancy Reference (September 2026)

FUNCTION / CONTRACT	PURPOSE
InayaBackupRegistry.sol	0x062c341aE4f11CB1dEa1B0D3930d52902F97f48a (BSC Testnet) — records redundancy commitment + health-state transitions only, never replica data
InayaKernel.Backup.getBackupStatus / getBackupHealth / getRedundancyStatus / getRecoveryStatus / requestRecovery	custody-sdk client, published to npm
Health states	PROTECTED / REBUILDING / DEGRADED / RECOVERY_REQUIRED / RECOVERY_FAILED — pure state machine, 19/19 unit tests
Pin-status check	Every 15 min, 3-strike grace window before flagging a replica
Integrity check	Daily, content-hash comparison, zero grace — catches real corruption immediately

Providers: Pinata + Filebase (S3-compatible, IPFS-backed, separate infrastructure/failure domain). Filebase gotcha: its S3 endpoint is s3.filebase.io, not .com; the AWS SDK's default auto-attached request checksum causes a generic AccessDenied against it unless disabled. Full writeup: docs/backup-redundancy-architecture.md.

SECTION 15

AI Guarded Execution & Audit Trail Reference

(September 2026)

FUNCTION	PURPOSE
proposeAiAction({orgId, assistantSurface, toolName, targetRecordType, targetRecordId, proposedAction, args, actorEmail, canPropose})	Inserts a PENDING_APPROVAL request; idempotencyKey = sha256(orgId+toolName+targetRecordId+args+hourBucket) dedupes same-hour repeats
reviewAiAction(..., canApprove)	Approve sets unlockAt = now + 36h (SETTLEMENT_DELAY_MS); reject is terminal
classifyRisk(targetRecordType, proposedAction)	Never returns undefined — unmatched combinations default MEDIUM; EXPENSE/INVOICE/PURCHASE_ORDER default HIGH
api/cron/execute-approved-ai-actions	The only code path that calls a real transitionX() for an AI-originated change, and only once unlockAt has passed; runs as a synthetic org-manager membership, attributed in the log to the human approver
appendAuditEntry({orgId, recordType, recordId, actorEmail, action, previousState, newState, metadata})	src/lib/auditChain.js — entryHash = sha256(prevHash + canonicalJSON(eventFields)); optimistic CAS on audit_chain_heads, retries on conflict, never corrupts under contention
verifyChainIntegrity(orgId)	Walks the full chain, recomputes every hash; returns {valid:false, brokenAtSeq, reason} on the first mismatch (gap/reorder, prevHash mismatch, or content altered after write)
GET /api/orgs/audit/export?orgId=&format=json csv	Org-scoped (owner/admin only) full-chain export — every field needed to independently recompute and verify, so a customer never has to trust a UI badge

9 gated domains (EXECUTORS map in ai-action-requests.js): TASK, EXPENSE, DOCUMENT, EMPLOYEE, INVOICE, LEAVE_REQUEST, PURCHASE_ORDER, PURCHASE_REQUEST, DEAL — one corresponding propose_* tool each in ai-business-tools.js. State machine: PENDING_APPROVAL → APPROVED → QUEUED → EXECUTED|EXPIRED, or PENDING_APPROVAL → REJECTED, or APPROVED → CANCELLED (only before unlockAt). 19 automated tests, 11 adversarial.

SECTION 16

Web3 App Store, NFT Vault & MFA Reference
(September 2026)

FUNCTION / ROUTE	PURPOSE
submitAppListing / listApprovedListings / listPendingListings / listListingsBySubmitter / reviewAppListing	src/lib/appStoreListings.js — full submission→threat-check→review→approve pipeline
POST /api/apps/submit, GET /api/apps/listings, GET /api/apps/my-listings	Developer-facing App Store API
GET /api/admin/app-store/pending, POST /api/admin/app-store/[slug]/review	Admin review queue, passphrase-gated same as /admin/audit
src/app/apps/embed/[slug]/page.js	Sandboxed iframe host for externally-hosted app submissions — no allow-same-origin + allow-scripts together
checkNftContractSupport / discoverOwnedTokens	src/lib/nftDiscovery.js — ERC-721 + Enumerable only, no indexer
signNftBackup	src/lib/nftBackupAuth.js — wallet-signature auth reused for NFT backup requests
POST /api/nft/backup, GET /api/nft/backups	NFT Vault backup API
src/components/AddressRiskCheck.js	Debounced live Security Layer threat-registry check, wired into the Bridge page's recipient field
MfaSettings.js / MfaVerifyScreen.js	TOTP + Firebase-Phone-Auth SMS MFA for Business Workspace login

SECTION 17

Sovereign Enterprise OS Reference (September 2026)

MODULE	KEY EXPORTS / FUNCTIONS
src/contexts/OrgContext.js	OrgProvider, useOrg() → {email, membership, orgId, role, orgs, switchOrg, logout, can: {manageOrg, accessDepartment, manageFinance, accessFinance, manageHR, accessHR, isDepartmentManager}}
src/contexts/WalletContext.js	WalletProvider, useWallet() → {walletAddress, walletBalance, isConnected, selectedWalletName, signMessage}
src/lib/orgGates.js	Client-safe extraction of orgs.js's 7 pure permission gates — re-exported from orgs.js under the same names for the ~60 existing server-side importers
src/lib/trustHealth.js	computeTrustHealthSnapshot(({scope:"org" "wallet", ...}) → {overallStatus, auditTrail security, aiActions backup, businessHealth, scopeNotes}
GET /api/orgs/trust-health, GET /api/wallet/trust-health	Org: requireMembership-gated. Wallet: unauthenticated, aggregate counts only
src/lib/notifications.js	createNotification (idempotent on dedupeKey), listNotificationsFor, markRead, markAllRead — collections: notifications, notification_reads
GET/POST /api/orgs/notifications/*, /api/wallet/notifications/*	Org: session-cookie auth. Wallet: GET unauthenticated, mutations signed via verifyMetadataAuth's existing message format
src/lib/orgSearch.js, src/lib/walletSearch.js	searchOrg(({orgId, membership, email, query}) over getAccessibleScope(); searchWallet(({walletAddress, query}) over metadata_files
src/components/CommandPalette.js	Cmd/Ctrl+K overlay, surface-agnostic — onSelect(result) left to the caller's own real navigation function
src/lib/activityCenter.js	generateWhatChanged(({scope, ..., period}) → {sections: [{module, bullets}]}); reuses business-brief.js's BRIEF_PERIODS
src/lib/ai-os-router.js	buildOsContext, getOsToolDeclarations(scope), runOsTool(name, args, ctx), osSystemInstruction — business_/security_ prefixed dispatch
POST /api/ai/os-chat, POST /api/ai/os-chat-wallet	Org: requireMembership-gated. Wallet: identityId passed directly, matching security-chat's existing precedent
src/components/business/OsHomeView.js, src/components/OsHomeSection.js	The composed OS Home screen per surface — TrustHealthCard, OS Assistant widget, What-Changed preview, module tiles, surfaced links
open_module_window(label, path)	Tauri command, both inaya-desktop and inaya-dapp-desktop's src-tauri/src/lib.rs — pops a module into its own native window

Every org-scoped route follows the exact ensureOrgIndexes()+requireMembership()+getAccessibleScope()-or-narrower pattern dashboard/route.js already established; every wallet-scoped GET follows the same unauthenticated-aggregate-only trust tier as the pre-existing list-files/backup-status routes. No existing route, collection, or component was renamed, removed, or had its behavior changed — additive only.

SECTION 18

Healthcare & Legal OS Reference (September 2026)

FUNCTION / ROUTE	PURPOSE
health-patients.js: createPatient, assignCareTeamMember, findDuplicatePatientCandidates, mergePatients	Patient registry; care-team assignment is the join row getAccessibleScope() reads for visibility
health-clinical-workflow.js: TRANSITIONS (draft→review→sign→lock), amendClinicalRecord	Locked records are never overwritten — amend creates a new linked record, marks the original AMENDED
health-consent-workflow.js, health-roi-workflow.js, health-breakglass.js	Consent record/withdraw; ROI request→authorize→review (hands off to export-center.js); break-glass = a real time-limited care-team assignment, not a bypass path
legal-matter-workflow.js: createMatter, assignMatterTeamMember; legal-conflict-workflow.js: searchConflicts, recordConflictCheck	Matter lifecycle + team assignment; conflict status is always potential/cleared/escalated, never a raw boolean
legal-evidence.js, legal-custody.js: recordCustodyEvent; legal-hold-workflow.js: createLegalHold, releaseLegalHold	Evidence metadata over an org_documents FK; every custody event chained via appendAuditEntry; holds gate retention.js's checkDispositionAllowed()
trust-accounting.js: recordDeposit, recordWithdrawal, getMatterTrustBalance	Atomic balance recheck before every withdrawal — no jurisdictional (IOLTA) compliance claimed
ai-health-tools.js, ai-legal-tools.js	buildXContext/X_TOOL_DECLARATIONS/runXTool/xSystemInstruction — read/summarize/draft only, wired into ai-os-router.js with health_/legal_ prefixes
industry-config.js: requireVertical(orgId, expectedVertical)	The door-lock — every Health/Legal route calls this right after requireMembership(); 403 on any mismatch, fail-closed as 'general' with no vertical set
25 routes under /api/orgs/health/*, /api/orgs/legal/*	Patients, consents, ROI, break-glass, appointments, billing, care-team, research (health); clients, prospects, matters, matter-team, conflict-checks, evidence, holds, discovery, redaction, deadlines, contracts, entities, trust-accounting, time-entries, billing (legal)
HealthView.js, LegalView.js (src/components/business/)	The only two modules with an actual screen today — registry/list + create + detail modal, mirroring CRMView.js's structure
src/lib/integrations/{health,legal}/*Adapter.js + adapterStub.js	13 documented third-party interfaces (FHIR, HL7, DICOM, lab, pharmacy, claims, e-filing, legal research, e-signature, email, calendar, accounting, SSO) — configured:false until real env credentials are present, never fabricated

Patient/matter visibility is assignment-based (health_care_team_assignments/legal_matter_team_assignments), a genuinely different model from every other Business Workspace module's department-scoping — wired directly into getAccessibleScope() rather than a parallel resolver. 20 domain modules have real backend + API; only 2 (Patients, Matters) have a screen. Two real bugs were caught and fixed during this build (break-glass had no expiry enforcement; 14 POST route handlers referenced an out-of-scope variable after the vertical-lock was added) — both now covered by permanent regression tests.

SECTION 19

Financial Services & Regulated Enterprise OS Reference (2026)

FUNCTION / ROUTE	PURPOSE
compliance-controls.js, compliance-evidence.js, control-testing.js	Control library + evidence vault; a failed control test auto-opens a Finding via the shared finding lifecycle
compliance-policies.js: createPolicyDraft→submitForReview→approve→publishPolicy, amendPolicy	Publish sets immutable:true; amendPolicy is the ONLY path forward from PUBLISHED, always inserting version+1, never mutating the published row
regulatory-examination-access.js: createExaminerMagicLink, exchangeMagicLink	Scoped (examinationId, requestId) access, deliberately its own token pattern (not a retrofit of the unrelated investor dataroom.js)
financial-entities.js, financial-funds.js: isFundTeamMember	Entity-scoped fund visibility — same assignment-based shape as isCareTeamMember/isMatterTeamMember
investment-committee.js, cap-table.js, board-management.js	IC decision lifecycle (finalized decisions are versioned, never overwritten); cap-table is an ingest/evidence layer, never a transactional share registry
privileged-access.js: requestElevation→approveElevation, grantBreakGlass, reviewSession	Cross-vertical from day one — the approver must be a different person than the requester; break-glass is ACTIVE immediately but always forces a post-event review
integrations.js: INTEGRATION_PROVIDERS, configureIntegration, recordSyncRun	A connection can only ever reach ACTIVE through a real recorded sync run — never through configuration alone
ai-audit-tools.js, ai-regulatory-tools.js + 7 more role-specific copilots	All 100% read/summarize/flag — a compliance-certification request is refused at the tool layer before any summarization runs
~20 API subtrees under /api/orgs/{regulated,financial,private-capital}/*	Every route: ensureOrgIndexes → requireMembership → requireVertical → lib call, statically verified by vertical-lock-wiring.test.mjs
RegulatedView.js, FinancialView.js (src/components/business/)	Full tab-router UI — Controls, Evidence, Findings, Policies, Risk, Audits, Examinations, Dashboard (Regulated); equivalent depth for Financial

All ten SOW phases are live-verified end-to-end on web: control creation/testing/finding lifecycle, the policy publish-immutability guard, an examiner magic-link's one-time-use cycle, entity-scoped fund/deal visibility, and the board-report/export-package immutability guards. Nothing on inaya-mobile for any of these three verticals yet. Every third-party integration (fund admin, custodian, KYC/AML, etc.) is a documented, honest not-configured stub.

SECTION 20

Government & Public Sector Sovereign OS Reference (2026)

FUNCTION / ROUTE	PURPOSE
citizen-records.js: createCitizenRecord, assignCitizenRecord, requireCitizenRecordAccess	requireCitizenRecordAccess() is the load-bearing gate — department membership or governmentRole:'staff' alone is never enough, only an explicit assignment (or org owner/admin) grants access to one specific record
government-cases.js: CASE_STATES (OPEN→ASSIGNED→IN_PROGRESS→PENDING_REVIEW→RESOLVED→CLOSED)	Same TRANSITIONS-map + atomic findOneAndUpdate template as incidents.js; a case linked to a citizen record inherits that record's need-to-know check
policy-knowledge-base.js: createEntryDraft→publishEntry, amendEntry	compliance-policies.js's exact publish-immutable lifecycle, recontextualized — no updateEntryDraft() path reachable once PUBLISHED
government-audit.js: logCitizenRecordAccess, logGovernmentDocumentRead	Government-vertical-only: every document READ is chain-logged, not just writes — an additive branch inside the existing retrieve route, gated so no other vertical's behavior changes
government-dashboard.js: getGovernmentDashboard	Operations + security readiness as two separate honest panels — a fresh org with zero audit-chain entries reports 'unknown,' never a fabricated 'valid'
ai-government-tools.js: search_citizen_records, summarize_case, get_dashboard_summary	100% read-only; need-to-know is re-checked INSIDE the tool per record, not just trusted from the API layer above it
privileged-access.js (reused, zero new code)	Break-glass access for Government orgs — already cross-vertical since §33, no government-specific fork needed
integrations.js: 5 new government_systems providers	Civil/national ID registry, legacy government ERP, GIS/land records, public records portal, interagency data exchange — all configured:false stubs
11 routes under /api/orgs/government/*	citizen-records (+assign), cases (+transition), policy-kb (+transition/publish/amend), dashboard — statically verified by the extended vertical-lock-wiring.test.mjs

FUNCTION / ROUTE	PURPOSE
GovernmentView.js (src/components/business/)	Tab-router UI — Citizen Records, Cases, Policy Knowledge Base, Dashboard — wired into the Business Workspace nav and org-creation flow, confirmed rendering live
51 new automated tests, including the two load-bearing properties written before their implementation: citizen-record access requires an actual assignment, and a published policy knowledge base entry can never be mutated in place. Nothing on inaya-mobile yet. No government certification, accreditation, FedRAMP authorization, or jurisdiction-specific compliance claim exists or is claimed anywhere.	
INAYA NETWORK	INAYA-DEV-2026-V1

SECTION 21

Institutional Trust Infrastructure Reference (September 2026)

FUNCTION / ROUTE	PURPOSE
evidence.js: getEvidenceTrail, verifyOrgEvidenceIntegrity, exportEvidencePackage	Read-only merge of listOrgActivityForRecord() + linked aiActionRequests into one timeline; integrity check is a pass-through to auditChain.js's verifyChainIntegrity()
org-trust.js: proposeTrustRelationship, respond, revokeTrustRelationship, isTrustedAccess	New orgTrustRelationships collection; accept/reject filtered by findOneAndUpdate({toOrgId: caller}) so an org can never approve its own proposal; isTrustedAccess is a query primitive no other module calls yet
POST/GET /api/orgs/trust-relationships, PATCH /api/orgs/trust-relationships/[relationshipId]	Session-authenticated cross-org trust CRUD
api-keys.js: generateToken/hashToken convention, requireApiKey	Raw key shown once at creation, only hash persisted; resolves to a synthetic owner-level membership permanently bound to the key's own orgId — no request parameter can override it
POST/GET /api/orgs/api-keys, DELETE .../[apiKeyId]	Session-authenticated key management, owner/admin only
GET /api/public/v1/audit/verify, /api/public/v1/evidence, /api/public/v1/permissions/check	API-key-authenticated (Bearer). evidence requires both recordType+recordId (no org-wide dump); permissions/check reports an org-level capability flag, not per-user RBAC
institutional-attention.js: getAttentionItems	AI-tool-only (get_attention_items in ai-os-router.js), not a REST route; merges pending-approvals (re-checked against the real approval gate), overdue tasks, pending documents
src/app/trust/page.js — /trust	Public, unauthenticated. Client-side verifier reimplements auditChain.js's hash formula via Web Crypto; auto-detects genesis vs. scoped exports (fixed in 9345df5) and reports the correct guarantee strength for each

Zero mainnet references across every file/route touched by this section, confirmed by direct grep. Every module ships a co-located test file (org-trust, evidence, institutional-attention, api-keys), none mocked against a fake database.

SECTION 22

Autonomous Resilience Layer Reference (September 2026)

FUNCTION / ROUTE	PURPOSE
resilience-policy.js: createPolicy, updatePolicy, evaluateRtoRpo	requiredRTOMinutes/RPOMinutes + critical-asset categories + test frequency; evaluateRtoRpo is a plain actual<=required comparison, no fuzzy scoring
resilience-canary.js: ensureResilienceHome, createCanaryAsset	Synthetic org-owned documents (isResilienceCanary:true, hidden from normal browsing) pushed through the real InayaKernel.disperseAndSlice() + real pinning-provider .pin() + real backupEngine.replicateShard() — never a mock of these functions
resilience-orchestrator.js: runResilienceTest, runScheduledResilienceTests	13-step test: real reconstruction via InayaKernel.reconstructAndDecrypt(), real getPinStatus() dependency check, real getDocumentAccessLevel() permission check, wall-clock RTO, replica-staleness-based RPO; atomic RUNNING-status duplicate-run guard
GET /api/cron/resilience-tests	CRON_SECRET Bearer-gated daily sweep, registered in vercel.json; per-policy try/catch so one broken policy never blocks the rest
resilience-status.js: deriveState	Pure function → Unknown / Test Due / Verified / Failed / Degraded, unit-tested directly; a PAUSED policy always reports Unknown regardless of test history
GET /api/orgs/resilience/status, POST /api/orgs/resilience/policies/[policyId]/run	Dashboard read + manual 'Run now' (canManageOrg-gated, triggeredBy:'manual' vs. cron's 'schedule')
ai-resilience-tools.js: get_resilience_status, get_last_test_result, list_failed_assets, explain_test_failure, trigger_resilience_test_now	First four are structurally read-only (no mutation code path exists); trigger_resilience_test_now can only start the same non-destructive test the cron runs, kept outside Guarded Execution because there is no destructive-workflow surface for it to gate

Two small, additive fixes to the pre-existing backupEngine.js's replicateShard() (expose providerRef; accept an explicit primaryProviderRef) were required and made in this same pass, both non-breaking. Verified live against real Filebase network calls during development.

SECTION 23

Node Operator Dashboard Reference (September 2026)

FUNCTION / ROUTE	PURPOSE
nodeOperatorAuth.js: loginWithWalletSignature, requireNodeSession, linkWalletToSession	New node_sessions collection, 12h cookie (inaya_node_session), reuses the existing verifyNodeAuth() signature check unchanged; fleet-linking requires the linked wallet's own signature
nodeUptimeHistory.js: getUptimeForWindow	Hourly-cron-populated node_uptime_snapshots; sets insufficientData:true and reports the real coverage window rather than backfilling or estimating history it doesn't have
nodeChainReads.js: getOnChainNodeInfo	Real ethers.JsonRpcProvider reads against the deployed node-registry contract (same address the existing settlement-release cron uses); tier/commission explicitly sourced on_chain since the off-chain Mongo tier field is set once and never revisited; fails honest-empty, never fabricates a tier
GET /api/nodes/operator/{me,uptime,qualification,rewards,events,network,fleet}, POST .../link, .../login, .../logout	network is the only unauthenticated route (public aggregate stats); every other route scopes strictly to the session's own wallet — no client-supplied node ID is ever trusted
GET /api/cron/nodes-snapshot	Bearer CRON_SECRET-gated hourly snapshot job, registered in vercel.json
src/app/operator/page.js + OperatorOverview/Uptime/Qualification/Rewards/Events/Network/Fleet.js	Seven-tab dashboard, standalone MetaMask connect-and-sign login independent of the dApp's WalletContext and Business Workspace's session

No claim/release control anywhere in the dashboard — settlement stays fully automated by the pre-existing timelock cron. Public /network reports off-chain tier distribution, not a live per-node chain read, an explicitly disclosed scalability trade-off. Tested with real ethers.Wallet signatures, treated internally as the highest-risk new surface in this pass.

SECTION 24

Four New Business Primitives Reference: Sign, DePIN Storage, Escrow, Attestation (September 2026)

FUNCTION / ROUTE	PURPOSE
signing-workflow.js: recordSignature, verifyInayaSignSignature, anchorSigningRequestOnChain, verifySigningRequest	Wallet path = real ethers.verifyMessage() ECDSA check, 5-min freshness window; session_consent path is explicitly labeled, never presented as equivalent; anchor is a real, optional transaction (batchRegisterAssets on the existing custody contract) over a completion-certificate hash, not a per-signature on-chain record
storage-manager.js: registerOrgStorageNode, getOrgStorageOverview, reportOrgStorageNodeHealth	New org-scoped control plane, confirmed unconnected to the public node-operator registry; eligibility is a static 'routing_not_yet_supported'; allocation/replica figures reused as-is from orgPlans.js + backupEngine.js, not recomputed
escrow-workflow.js: transitionEscrow, confirmMilestone, releaseMilestonePayment	State machine only flips a status field; the ONLY function moving money is releaseMilestonePayment, reachable only after a proposeAiAction() Guarded Execution approval plus the standard 36h SETTLEMENT_DELAY_MS; writes a real payments row, re-validates dispute status at execution time
financial-attestation.js: buildCanonicalDataset, generateAttestation, verifyAttestation	sha256 hash-commitment over real permission-scoped, currency-converted financial records; only revenue_threshold/expense_threshold implemented, solvency_threshold explicitly rejected (no assets/liabilities data model exists); verification independently re-runs the query and recomputes the commitment
POST/GET routes under /api/orgs/sign, /api/orgs/escrow, /api/orgs/storage, /api/orgs/attestations (~19 routes)	SignInView, EscrowView, StorageManagerView, AttestationsView in src/components/business/ — real nav entries in page.js's Operations group, real org-scoped API-backed screens

Each module's own header comment states plainly what's real vs. not: Escrow is never described as on-chain or non-custodial anywhere in its code, routes, or UI; Attestation names the specific ZK libraries confirmed absent from the full dependency tree to avoid that label being misapplied; DePIN Storage Manager's region preferences are always labeled Declared, never Active. A real nested-object canonicalization bug found while testing Attestation was also fixed in Sign's anchor certificate, which shared the same flaw. Test coverage: inaya-sign, milestone-escrow, depin-storage-manager, financial-attestation.

SECTION 25

Business Workspace Operational Enhancements & Guided AI Assistant Reference (September 2026)

FUNCTION / ROUTE	PURPOSE
finance/invoices/[invoiceId]/pdf/route.js	Returns print-styled HTML, not a binary PDF — the route's own comment confirms no PDF library exists in the app; values pulled verbatim from the invoice record, never recalculated
currency.js: convert, SUPPORTED_CURRENCIES, USD_RATES	Static, explicitly dated reference-rate table (RATES_AS_OF), not a live FX feed by design; returns an error rather than guessing on unsupported pairs
inventory.js: transferStock, recordStockMovement	Two ledger movements + a materialized stock_levels update sharing one transferId; a real compensating ADJUSTMENT reversal if the second leg fails after the first commits
purchase-order-workflow.js: updatePurchaseOrderItems	Atomic findOneAndUpdate({status:'DRAFT'}) guard — editing outside DRAFT returns a real 409; reruns the same validator the create route uses
procurement/orders/[orderId]/email, procurement/reports	Real supplier email with real line items via the existing sendEmail() primitive (documented no-op if RESEND_API_KEY unset); 5 real report types incl. CSV export
insights route.js + business-insights.js: computeBusinessInsights	Real custom start/end date range on top of preset windows, validated (endDate<startDate → 400), computed via the same permission-scoped getAccessibleScope() every other route uses
business-chat/route.js — Gemini (gemini-3.5-flash-lite) via @google/genai, Groq fallback	Real, production-hardened LLM call: per-call timeout, request-level safety budget, retry-on-503 — added after an observed Gemini-queuing production issue
guided-workflow-catalog.js (static data, no LLM calls) + guided-tasks.js (state machine)	~10 hardcoded workflows; the model only ever relays pre-authored instruction text verbatim — structurally prevented from inventing step text; the task's own bookkeeping collection is the only thing it ever writes to
GuidedTaskPanel.js + inaya:guided-nav / inaya:guided-po-created custom events	Step advancement fires from real navigation events, real UI success events, or a manual 'I did this' confirmation — never the model's own assertion that a step is done

Creating a new business record (contact, deal, PO, document) has no automated tool path at all — the assistant can only guide a human through the real UI. Any actual mutation it can propose still routes through the pre-existing Guarded Execution approval-plus-36-hour-delay pipeline described in §26/§31. No new mutation surface was added by the guided-task feature itself.

SECTION 26

Multi-Cloud Storage Compatibility & Enterprise Storage Governance Reference (September 2026)

FILE / ROUTE	PURPOSE
s3-compatible/sigv4.js: verifySigV4Request, parseAuthorizationHeader	Real AWS SigV4 verification — canonical request → string-to-sign → derived signing key (HMAC chain) → signature; 15-min clock-skew window; UNSIGNED-PAYLOAD support
s3-compatible/azureAuth.js: verifySharedKeyRequest, s3-compatible/azureAuthMiddleware.js: authenticateAzureRequest	Azure Shared Key verification + real Microsoft Entra ID Bearer-token path (live Graph verification via integrationProviders/microsoft.js, mapped to org_members by email)
s3-compatible/credentials.js: issueS3Credential, checkScope, normalizeScope	Credential issuance with optional scope {bucket, prefix, operations, expiresAt}; checkScope() is the single enforcement function, called centrally in auth.js/azureAuthMiddleware.js after signature verification
s3-compatible/store.js, walletStore.js: putS3Object, deleteS3Object, assertNotProtected	Org- and wallet-side translation layers over org_documents/metadata_files; assertNotProtected() is the one chokepoint both the overwrite path and the delete path call through for Object Lock/Legal Hold enforcement
s3-compatible/store.js: putBucketVersioning, listObjectVersions, restoreObjectVersion	Bucket-level versioningStatus (Unversioned→Enabled→Suspended, one-way); restore copies an old version's bytes forward as a new current version rather than resurrecting the old version id
s3-compatible/store.js: putObjectRetention, putObjectLegalHold, putLifecyclePolicy, runLifecycleEnforcement	Object Lock (requires Versioning Enabled first; retention can extend, never shorten), Legal Hold, and per-bucket lifecycle rules with real enforcement that skips locked/held objects
api/s3/**/route.js, api/azure/**/route.js	Full S3 and Azure REST surfaces, including real sub-resources (?versioning, ?object-lock, ?versions, ?legal-hold, ?retention, ?versionId=, Put Block/Put Block List)
api/cron/s3-lifecycle/route.js	CRON_SECRET-gated route calling runLifecycleEnforcement() — same auth convention as api/cron/nodes-snapshot; registered in vercel.json's crons array
api/orgs/s3-compatible/manage/route.js, S3CompatView.js	Session-authenticated Business Workspace management API + console — separate from the SigV4-signed protocol surface, so Business Workspace never needs to hold/sign with an S3 credential
backupEngine.js: replicateShard (now called from putS3Object)	Closes a real, previously-existing gap: S3/Azure-compatible objects were pinned but never registered with the existing check-pins/verify-integrity/recovery crons
inaya-drive-helper/src/main.rs, sigv4.rs, s3client.rs	Standalone Rust crate (GPL-3.0, isolated on purpose) implementing WinFSP's FileSystemContext trait over a Rust port of the SigV4 signer, backed by the real /api/s3 endpoint
inaya-desktop/src-tauri/src/lib.rs: mount_inaya_drive, unmount_inaya_drive	Tauri commands that only spawn/kill inaya-drive-helper.exe as a child process — no GPL dependency in inaya-desktop's own Cargo.toml

Test coverage: test/s3-compatible-capabilities.test.mjs (19 tests: scope enforcement, versioning, Object Lock, Legal Hold, lifecycle, health registration) plus the pre-existing test/s3-compatible-sigv4.test.mjs (9) and test/s3-compatible-store.test.mjs (9) — all passing, zero regressions. Full writeup: docs/storj-inspired-storage-capability-expansion-report.md and docs/multi-cloud-storage-compatibility-report.md.

SECTION 27

Storage Interoperability Expansion Reference — Empty Folders, Migration Tooling, Signed URLs & GCS Auth (September 2026)

FILE / ROUTE	PURPOSE
inaya-drive-core/src/lib.rs, s3client.rs, sigv4.rs	New standalone Rust crate extracted out of inaya-drive-helper: the SigV4-signed S3 client and signer, with zero Windows/WinFSP dependency, now linked unmodified by both inaya-drive-helper (Windows/WinFSP, GPL-3.0) and the new inaya-drive-helper-linux (Linux/FUSE via the MIT-licensed fuser crate, UNLICENSED) — one real client implementation, not a per-platform rewrite. Each platform's own filesystem-trait binding (WinFSP's vs. FUSE's) stays separate, since the two OS filesystem APIs aren't interchangeable
s3-compatible/store.js: createS3Folder, getS3FolderInfo, deleteS3Folder, renameS3Folder	A real, durable folder record in a new s3_folders collection (mirroring the pre-existing Business Workspace metadata_folders shape) — a folder with zero objects is now listable via ?prefix&delimiter, where previously only shared object-key prefixes existed and an empty folder was invisible
inaya-migration-agent/bin/inaya-migrate.mjs	Standalone Node CLI: --source aws azure gcs, resumable via a local --manifest file (skips already-verified keys on rerun), --dry-run, --objects for a scoped subset; every source/destination credential stays local to the invoking machine, never transmitted to or stored by Inaya
s3-compatible/signedUrl.js: createSignedUrl, verifySignedUrl	Query-string presigned GET URLs (X-Amz-Expires/X-Amz-Signature convention): reuses the existing SigV4 signing key derivation rather than a new secret, rejects on expiry, method mismatch, or any tampered query parameter
googleAuth.js: verifyGoogleIdToken, s3-compatible/auth.js (Google ID-token branch)	Google Sign-In accepted as a direct auth path on the S3-compatible endpoint: verifies the Google ID token's signature and audience server-side, then resolves it to an existing org_members row by verified email — no separate Google-only permission tier or credential type
middleware.js: bucket-from-hostname rewrite, gated by S3_COMPAT_VIRTUAL_HOST_BASE	Virtual-hosted-style addressing (bucket-name.base-host/key, matching real AWS/GCS convention) — the Host header is parsed and rewritten to the existing path-style route before any other S3/Azure code runs, so the protocol surface itself is unchanged; unset (off) by default
evidenceExporter.js: buildEvidencePackage, canonicalizeForExport	Reads an org's own existing audit chain and storage-protection settings (Object Lock, versioning, retention) into one canonicalized, hash-stamped package — no new data is generated or claimed, only exported
evidencePdf.js: renderEvidencePdf	Deterministic PDF rendering of the same package for a non-technical reader, hash printed on the document itself so a paper copy can still be checked against the JSON export

Test coverage: test/s3-compatible-empty-folder.test.mjs (15 tests: placeholder-key creation/listing/deletion, cross-platform key-prefix handling, mount-restart persistence), test/signed-url.test.mjs (9: expiry, tamper rejection, method restriction, real wall-clock expiration — not a simulated clock), test/evidence-exporter.test.mjs (5), test/migration.test.mjs (6) — all passing against the real dev database, zero regressions in the pre-existing S3/Azure-compatible suites.

Two corrected findings, kept honest rather than restated as always-true: Google's own "gcloud storage" CLI was earlier reported unable to reach a third-party endpoint like Inaya's — it can, through an official (Google-labeled "unstable") configuration option, verified with real uploads and downloads. The older "gsutil" tool still cannot, due to a genuine bug in Google's own software; a workaround narrowed it further but did not fully resolve it. Remaining disclosed limitation: macOS Drive support is written but not yet compiled or tested on real Mac hardware. Full writeup: docs/inaya-drive-empty-folder-creation-report.md, docs/enterprise-adoption-and-market-reach-report.md, and docs/gcs-compatibility-extension-report.md.

SECTION 28

AWS S3-Inspired Storage Feature Expansion Reference (September 2026)

FILE / ROUTE	PURPOSE
s3-compat/store.js, walletStore.js: putObjectTagging, getObjectTagging, deleteObjectTagging	Real S3 tag semantics enforced server-side (10-tag max, 128/256-char key/value limits), stored as an additive `tags` field on the existing object document, never a second metadata collection
s3-compat/store.js, walletStore.js: contentSha256 field on putS3Object	The real, unsalted SHA-256 of uploaded bytes, computed and stored alongside the pre-existing salted fileHash (salted for an unrelated legacy dedup reason and therefore not independently client-verifiable) — exposed as x-amz-checksum-sha256 on GET/HEAD/PUT
Object route ([...key]/route.js): ? tagging (PUT/GET/DELETE), x-amz-tagging header	Real S3 PutObjectTagging/GetObjectTagging/DeleteObjectTagging sub-resource dispatch, plus tag-at-upload-time support via the standard x-amz-tagging query-string-style header
New: s3-compat/inventory.js: buildStorageInventory, renderInventoryCsv	Org-scoped JSON/CSV export of every live object (bucket, key, size, tags, checksum, version/lock state), computed from existing org_documents records only
New: s3-compat/batchOperations.js: runBatchOperation	SET_TAGS/SET_RETENTION/SET_LEGAL_HOLD across up to 1,000 keys, implemented as a loop over the existing per-object primitives with per-key success/failure reporting — no new low-level mutation path, Object Lock/Legal Hold enforced identically to a single manual request
New: s3-compat/analytics.js: computeS3BucketAnalytics	Real per-bucket object count/total-size/average-size/largest-objects/version-count/locked-count/legal-hold-count/tag-distribution, computed live, no materialized metrics table
New: s3-compat/policyAnalyzer.js: analyzeS3CredentialPolicies	Read-only credential-scope analysis (UNRESTRICTED_CREDENTIAL, NO_EXPIRY, BROAD_DESTRUCTIVE_SCOPE, BROAD_UNSCOPED findings) over the existing s3_credentials collection via listS3Credentials — zero writes; unusedCredentials is explicitly null (not []), since no lastUsedAt field exists to support that claim
Fixed: pinningProviders/pinata.js: authHeaders/resolveJwt	Detects a usable Pinata credential by shape (a JWT is three dot-separated base64url segments starting with "eyJ") across PINATA_JWT, PINATA_SECRET_API_KEY, and PINATA_API_KEY rather than one hardcoded variable name — this environment's real credential was a scoped-key JWT stored under PINATA_SECRET_API_KEY, previously unrecognized
Test coverage	test/s3-compat-expansion.test.mjs (9 tests: checksum correctness, tag validation limits, mixed-success batch jobs, Object Lock bypass rejection inside a batch job, org-scoped inventory isolation, per-bucket analytics correctness, policy-analyzer read-only guarantee) — all passing, plus the full pre-existing 57-test S3-compat suite (store/capabilities/empty-folder/sigv4) re-confirmed clean

Explicitly out of scope for this pass, per the SOW's own audit-first instruction not to build a feature merely because AWS has it: event notifications/webhooks (zero existing outbound-webhook infrastructure anywhere in this codebase to build on — only inbound receivers for Stripe/referrals/KYC exist); MFA/step-up authentication (Object Lock/Legal Hold already provide real, server-enforced protection against the highest-risk destructive action); Block Public Access (confirmed not applicable — the S3-compat auth layer has exactly three authenticated paths and a hard rejection otherwise, so there is no unauthenticated read path to block). Full writeup: docs/aws-s3-feature-expansion-report.md.

SECTION 29

Evidence Graph & Digital Twin Reference — Business Events, Passports, and Dependency-Graph Simulation (September 2026)

FILE / ROUTE	PURPOSE
New: src/lib/businessEvents.js: createBusinessEvent, getBusinessEvent, listBusinessEvents, addBusinessEventRelationship, getBusinessEventTimeline	References an existing invoice/PO/PR/AI-action-request by {subjectType, subjectId}, never copies it; department-scoped (or org-manager-only for AI-action subjects with no resolvable department) via the same canAccessDepartment/canManageOrg gates every other Business Operations record uses; timeline merges real org_activity entries from the event, its subject, and every linked relationship target
New: src/lib/businessEventExplain.js: explainBusinessEvent	Why? explainability — resolves each relationship target into INCLUDED/RESTRICTED/REFERENCED/UNAVAILABLE, evaluates one real, disclosed amount-threshold rule, and surfaces only the structured fields an AI action request actually stores (proposedAction, riskLevel, requestedContextSummary) — never internal reasoning, since none is persisted
New: src/lib/businessEventPassport.js: buildBusinessEventPassport, verifyBusinessEventPassport, renderBusinessEventPassportPdf	Reuses evidenceExporter.js's exact canonicalizeForExport hashing convention for the manifest hash; verification independently recomputes the hash and live-reverifies the org's audit chain, returning VERIFIED/INVALID/INCOMPLETE/UNKNOWN
New: src/lib/businessEventSimulate.js: simulateBusinessEventDecision	Re-derives PO_TRANSITIONS/PR_TRANSITIONS legality and the real canManageOrg/canAccessDepartment authorization checks without ever importing transitionPurchaseOrder/transitionPurchaseRequest/reviewAiAction — zero write path by construction
New: src/lib/digitalTwin.js: resolveDependents, traverseDependencyGraph	Computes the dependency graph on demand from real foreign keys already on real records (supplierId on a PO, departmentId everywhere, projectId on tasks/deals, assigneeEmail on tasks) — depth-bounded (default 3) breadth-first traversal, RESTRICTED nodes included but never traversed further
New: src/lib/digitalTwinSimulate.js: simulateDigitalTwinScenario	SUPPLIER_UNAVAILABLE, EMPLOYEE_ACCESS_REMOVED, PROJECT_DELAYED, WAREHOUSE_UNAVAILABLE — each traverses the real dependency graph and reports a numeric consequence only when a real stored field backs it (e.g. a task's own dueDate); every other effect is explicit UNKNOWN with a stated reason
API	/api/orgs/business-events/** (create/list/detail/relationships/why/passport/verify/simulate), POST /api/orgs/digital-twin/simulate — all session-authenticated via the existing requireMembership pattern
Test coverage	47 tests across test/business-events.test.mjs, test/business-event-passport.test.mjs, test/business-event-simulate.test.mjs; 9 tests in test/digital-twin.test.mjs — all passing against the real database, zero regressions to document-permissions.test.mjs or trust-health-v2.test.mjs

A real permission gap found and fixed during development. stockLevels rows have no departmentId of their own (see inventory.js) — the generic per-dependent-doc permission check that correctly gates every other entity type in digitalTwin.js would have silently defaulted a stockLevel row to visible regardless of the caller's actual department access, since there was no departmentId field on the doc itself to check. Fixed by gating simulateWarehouseUnavailable() on the WAREHOUSE entity's own departmentId before any traversal begins, rather than relying on the generic per-dependent check; documented directly on resolveDependents() so a future caller doesn't reintroduce the same gap by exposing a raw "traverse anything" endpoint without the same starting-point check.

Cross-organization Digital Twins were evaluated in a dedicated feasibility pass (docs/digital-twin-cross-org-privacy-feasibility.md) covering zero-knowledge proofs, secure multiparty computation, trusted execution, cryptographic commitments, and a scoped purpose-bound API — recommendation is to defer until a real, named cross-org query use case exists, reusing the pre-existing orgTrustRelationships collection and the existing audit chain when that need arrives, rather than building ZK/MPC/TEE infrastructure speculatively. Twin snapshots for reproducibility, a graph-visualization UI, and natural-language scenario creation are likewise deferred. Full writeup: docs/evidence-graph-business-event-layer-report.md, docs/digital-twin-simulation-layer-report.md, docs/digital-twin-cross-org-privacy-feasibility.md.